

Adult Income Census Semester Project

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Introduction:

The purpose of this project is to use machine learning techniques to predict whether an individual earns more than \$50,000 per year based on demographic and employment-related information. For this analysis, I used the Adult Census Income dataset from the UCI Machine Learning Repository. This dataset is commonly used for classification problems. It is well suited for this course because it contains a mix of numeric and categorical variables, along with a clearly defined binary target variable. The main goal of the project was not only to build a predictive model, but also to understand the patterns within the data and evaluate how different models perform on the same problem. Throughout the project, I employed the full machine learning process: data exploration, data cleaning and preprocessing, model training, model evaluation and comparison, and result interpretation using quantitative analysis and visualization.

Dataset Overview:

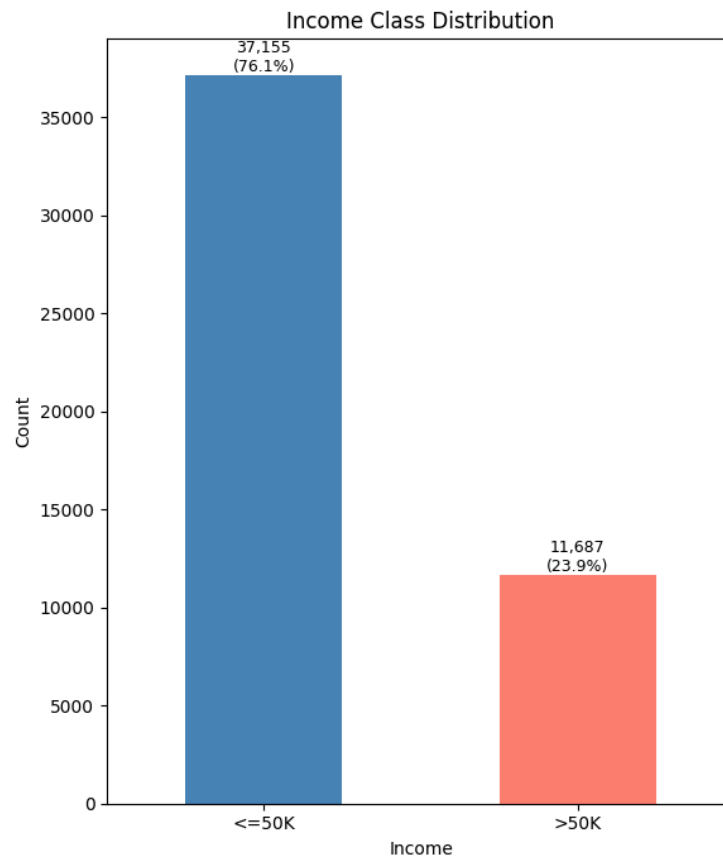
The Adult Census Income dataset was loaded using the `ucimlrepo` package, which provides direct access to datasets from the UCI Machine Learning Repository. After loading the data, the first output I reviewed was the dataset shape. The printed output showed that the dataset contains 48,842 rows, confirming there were plenty of observations to work with and that it easily met the project requirement of at least 1,000 records. Printing the first few rows of the dataset helped confirm that the data loaded correctly and gave an initial sense of what each feature represented. The dataset includes variables such as age, workclass, education, occupation, hours worked per week, capital gains, and capital losses. The target variable, income, indicates whether a person earns more than \$50K per year. Seeing both numeric and categorical features at this stage made it clear that preprocessing would be an important step later on.

Missing Value Analysis and Class Distribution:

The very first thing that I looked into was whether there were any missing values in the dataset. As shown by `df.isnull().sum()`, all numeric columns have no missing values. On the other hand, a few categorical columns such as workclass, occupation, and native-country had missing values. This was one of the reasons why I decided to fill missing values with the mode during preprocessing. Furthermore, I examined the income class distribution by displaying the value counts of the income column. The output shows that many more people earn \$50K or less than those who earn more than \$50K.

Income Class Distribution:

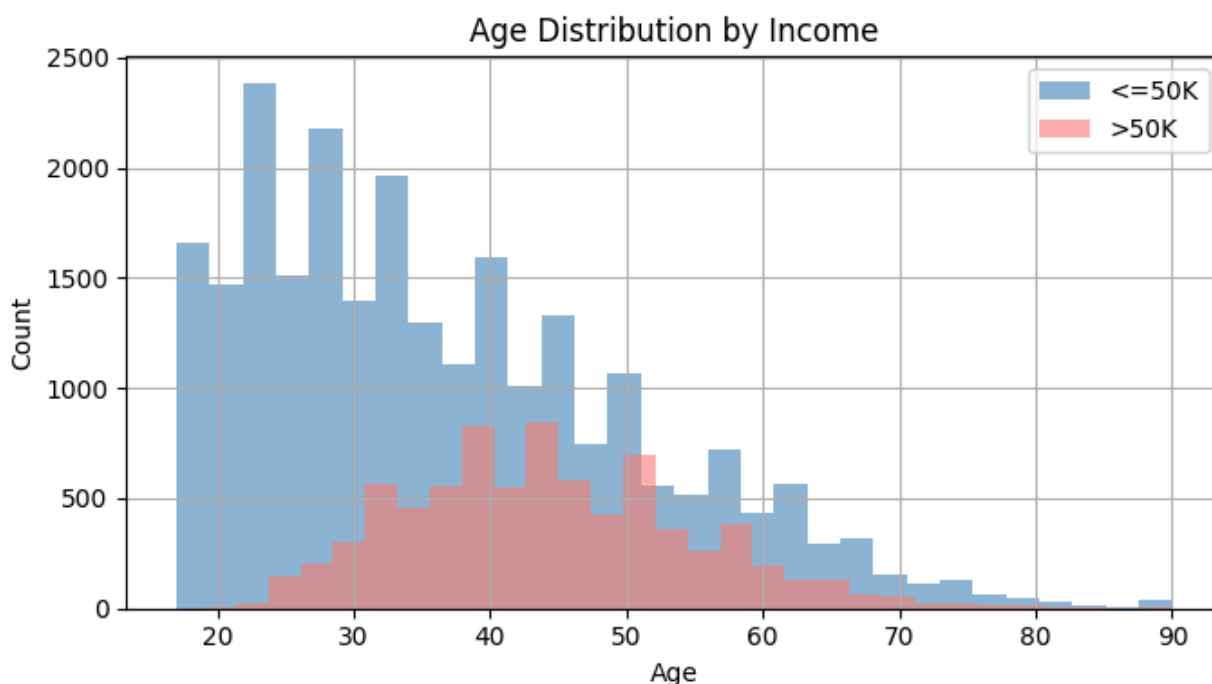
The first chart created was a bar graph illustrating the distribution of the income categories. The bar chart visually emphasized the output numbers by confirming that the “ $\leq 50K$ ” category accounts for the majority of the data. This type of visual analysis helps determine the initial challenge in predicting incomes. Predicting that everyone in the database earns no more than \$50K will yield high accuracy but will not identify those who earn more than \$50K. This is one of the reasons ROC-AUC was chosen as the metric for comparing the classifiers' effectiveness.



Age Distribution by Income Level:

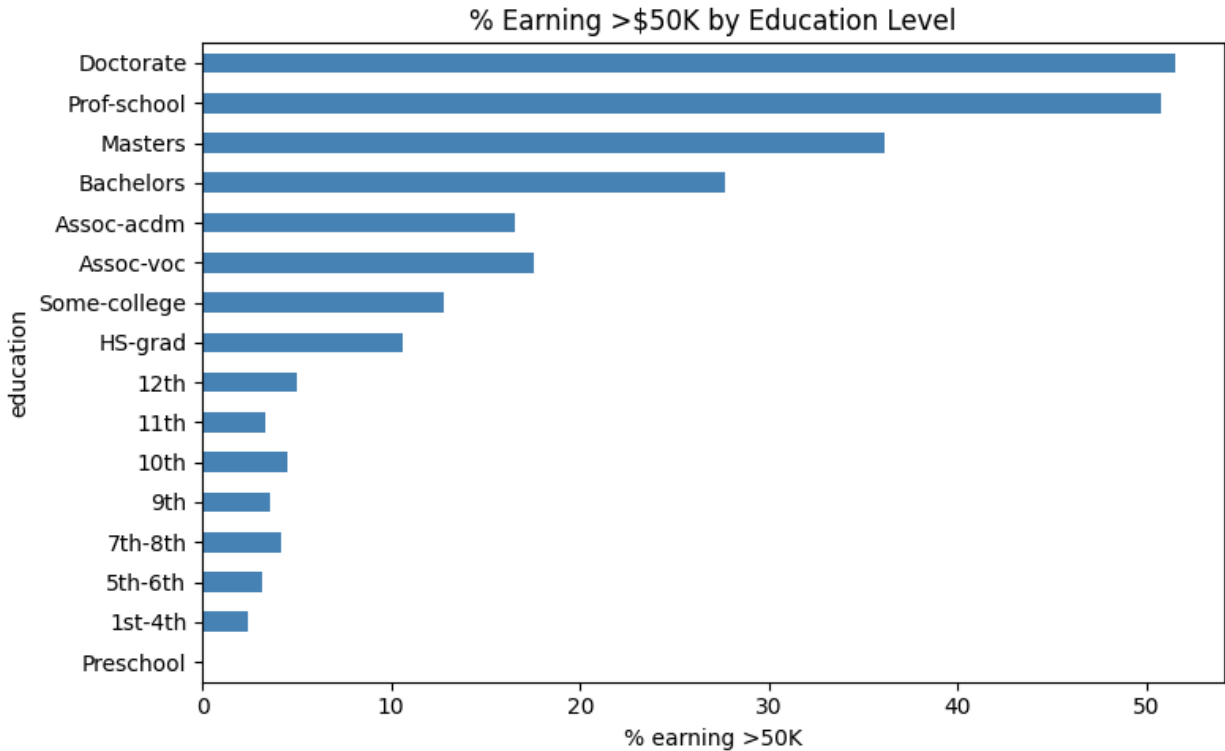
Another visualization focused on how age differs between income groups. The overlaid histograms showed that individuals earning more than \$50K tend to be older, on average, than those earning less than \$50K. The distribution for higher earners peaked at older ages, while the lower-income group had a much larger concentration of younger individuals. This plot suggested that age is an important predictor of income, but also showed significant overlap between the two

groups. This indicates that age alone cannot determine income level, which is why combining multiple features in machine learning models is necessary.



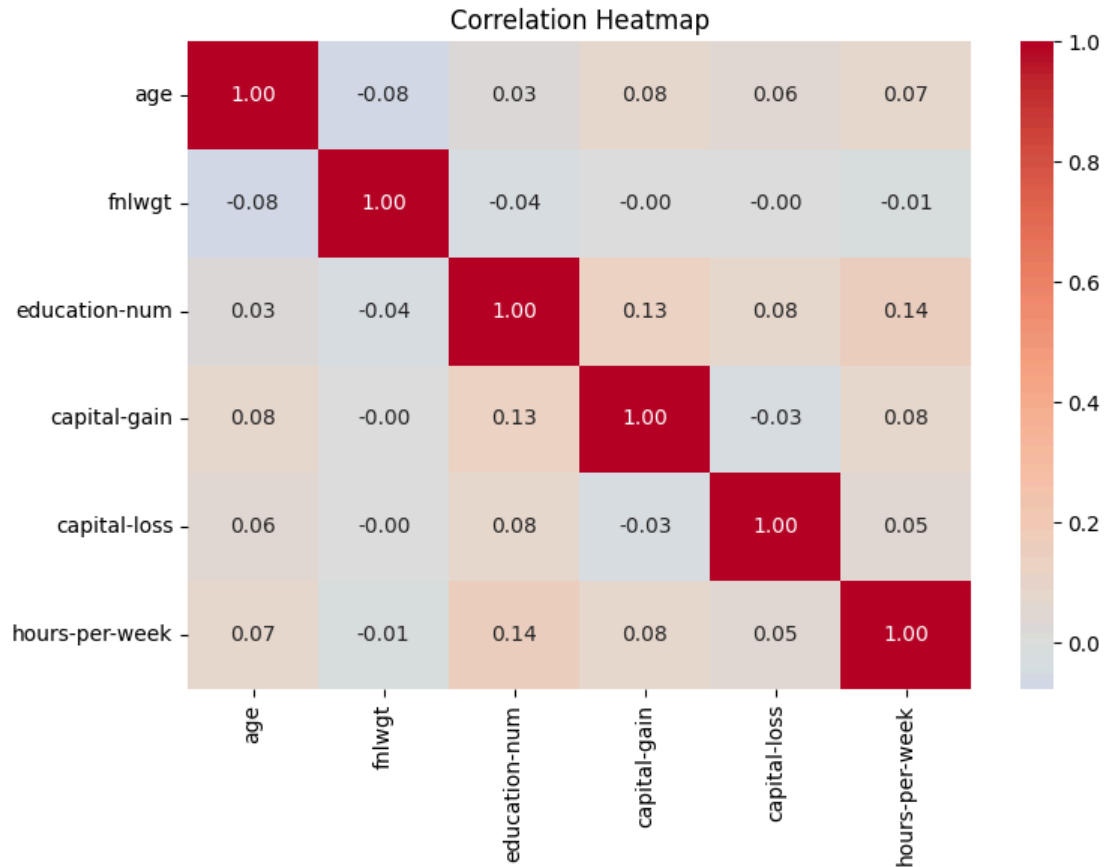
Education Level vs. Income Rate:

One of the most interesting visualizations was the horizontal bar chart showing the percentage of individuals earning more than \$50K for each education level. The trend was very clear: higher levels of education were strongly associated with higher income. Individuals with professional degrees, doctorates, and master's degrees had the highest rates of earning over \$50K, while those with less than a high school education rarely exceeded that threshold. This result was not surprising, but it was still helpful to see how strong the relationship was visually. Because of this, keeping the numeric education variable made sense for modeling, while the text-based education category could safely be removed to reduce redundancy.



Correlation Heatmap:

The correlation heatmap illustrated the connection between numeric variables. The most notable among these was the strong positive correlation between the capital gain variable and the income-related variables, as well as with years of education and age. On the other hand, capital loss and hours per week showed weaker correlation but can be considered, along with other factors, as useful variables. It helped validate the assumption that the determinant of income is the product of several factors, including financial, educational, and demographic aspects, instead of a single factor. Correlation does not necessarily equate to importance, but it served as good guidance before modeling.



Data Preprocessing and Feature Engineering:

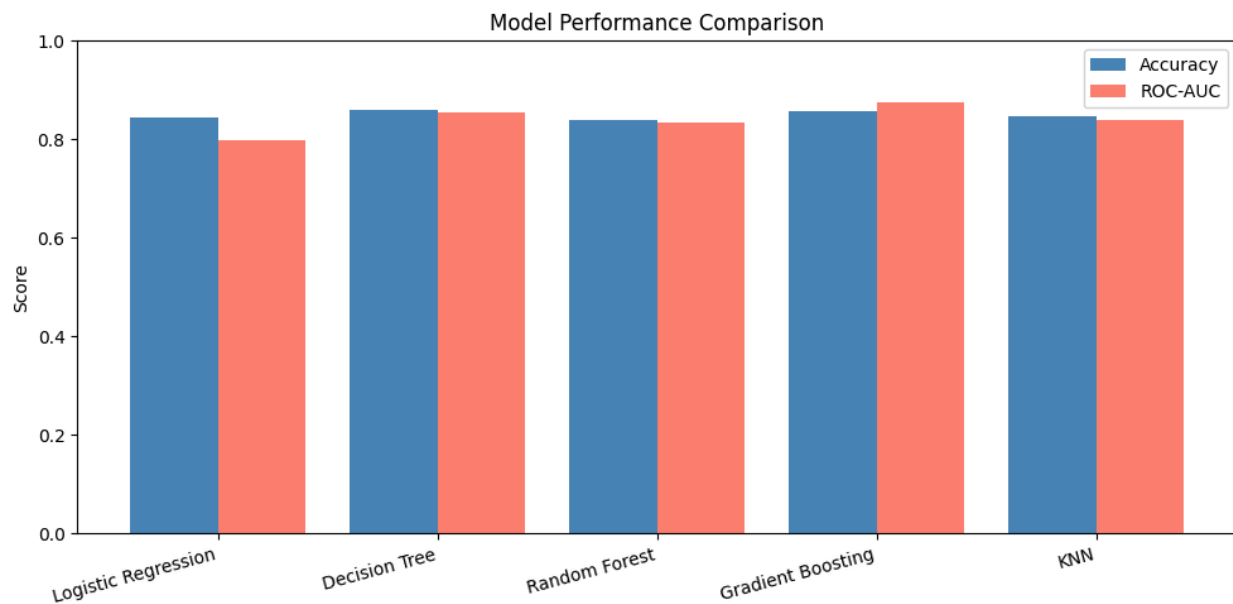
Before modeling, various preprocessing procedures were carried out. First, the income variable was manually recoded into a binary numeric form, where 1 indicated income above \$50K and 0 indicated income at or below \$50K. The purpose of the operation was to make the target variable suitable for classification models. To address missing values in categorical variables, the mode of each feature was used as the imputation value; this was done safely because the proportion of missing data was low. The fnlwgt variable was excluded from analysis because it was a census weighting indicator. Additionally, the education column was no longer needed because the numeric education column already contained all the information. After that, LabelEncoder was used to convert all the other categorical columns to numeric columns. I implemented an 80/20 train/test split with stratification to maintain class proportions. Since Logistic Regression and KNN are sensitive to feature scaling due to their reliance on distance measures and optimization, a StandardScaler transformation was applied.

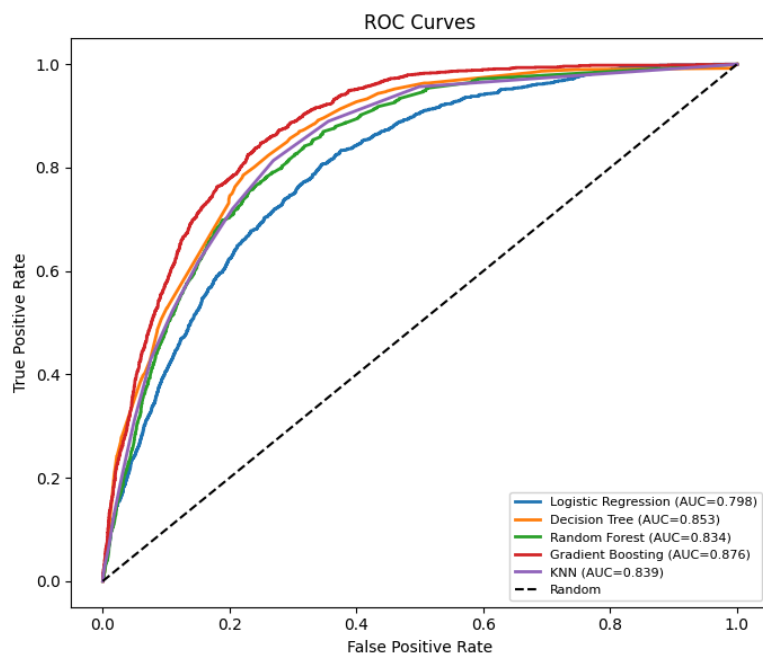
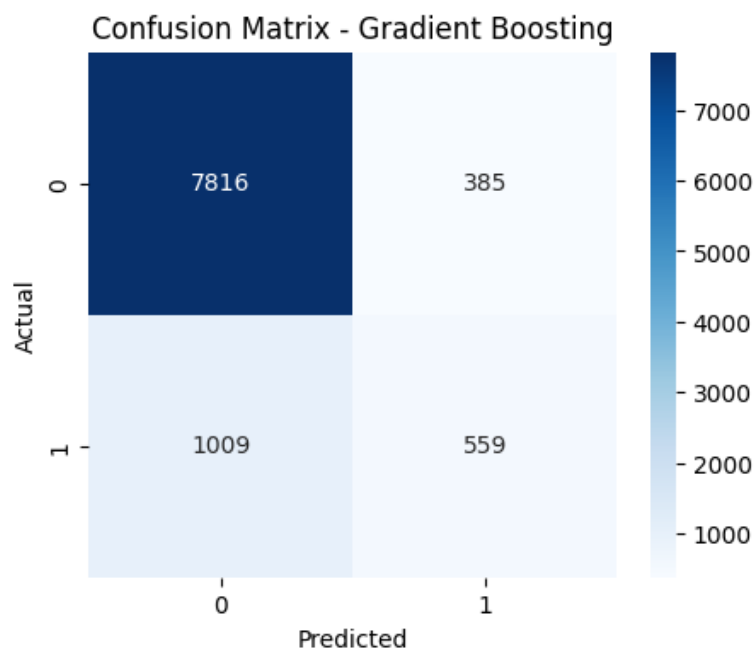
Model Training and Performance Evaluation:

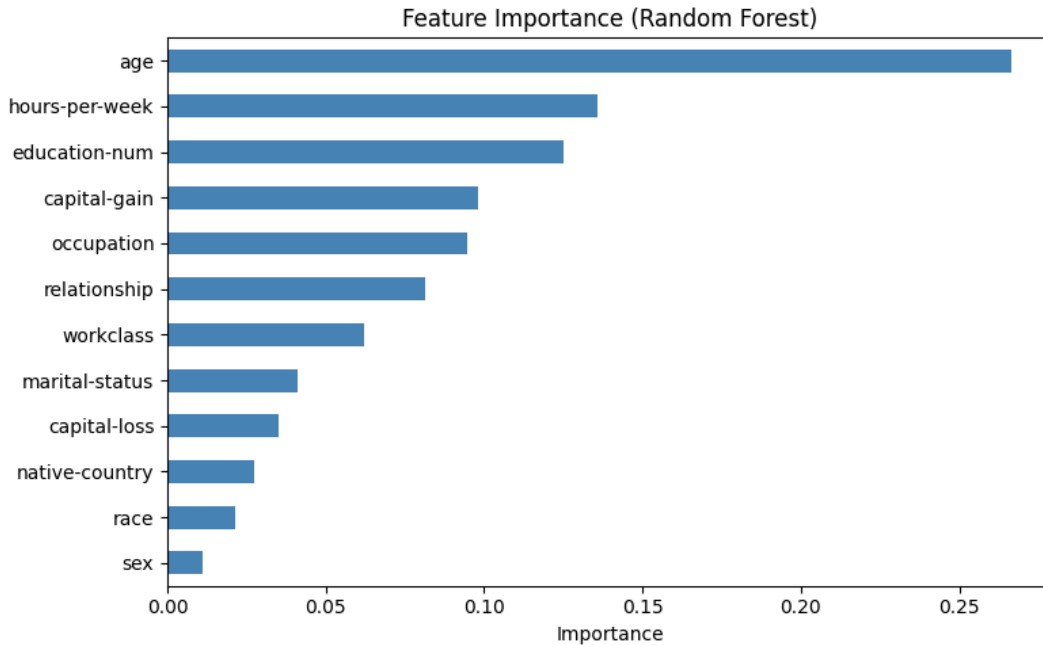
Five different classification models were trained: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and KNN. Training multiple models made it easier to compare how different approaches handle the same dataset. Based on the printed performance results, ensemble models such as Random Forest and Gradient Boosting performed best overall. They achieved the highest accuracy and ROC-AUC scores, which indicates strong predictive ability across different decision thresholds. Logistic Regression performed reasonably well as a simpler model, while KNN showed slightly lower performance, possibly due to the dataset's high dimensionality.

Model Comparison & ROC Curves:

The graph comparing accuracy and ROC-AUC scores for different models provided a clear visual comparison of their performance. Ensemble models performed better than other models because they can perhaps find the best possible way to fit the data. The ROC-AUC values provided further evidence for this conclusion. As one can see, ROC values for the Random Forest and Gradient Boosting models lie closer to the upper-left corner of the coordinate system. Models with weaker performance had curves closer to the diagonal line.







Feature Importance & Insights:

According to the feature importance plot obtained for the Random Forest algorithm, the most important features include Capital Gain, Number of Education, Age, and Hours Per Week worked. The above findings align well with those from the Exploratory Data Analysis.

Conclusion & Future Work:

Overall, this project successfully demonstrated how machine learning can predict income levels from census data. The exploratory analysis provided important insights, while data preprocessing helped improve data quality. On top of that, comparing different models showed that ensemble models perform especially well for classification problems like this one. For future improvements, we can tune hyperparameters, apply cross-validation, use encoding algorithms, and more.

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